Questions: Solving first-order linear ODEs using integrating factors

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Summary

A selection of questions for the study guide on solving first-order linear ODEs using integrating factors.

Before attempting these questions, it is highly recommended that you read Guide: Introduction to solving first-order linear ODEs using integrating factors.

Q1

Rearrange each first-order linear ODE below to be in the form, $\frac{dy}{dx} + P(x)y = Q(x)$ with respect to the variables in each equation.

1.1. $-\frac{\mathrm{d}y}{\mathrm{d}x} = y + 8.$

1.2. $\frac{1}{2x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}x}+5x}{y}.$

1.3. $\frac{\mathrm{d}x}{\mathrm{d}t} - \cos(t) = -x\cos(t).$

1.4. $x\frac{\mathrm{d}y}{\mathrm{d}x} - x^2 = y.$

Q2

Find the integrating factor of every question in question 2.

Q3

Solve each equation in question 2 for the general solution.

Q4

4.1. Find the exact solution to question 1.1. given the initial condition $y(0) = 0$.

4.2. Find the exact solution to question 1.2. given the initial condition $y(1) = 6$.

4.3. Find the exact solution to question 1.3. given the initial condition $x(0) = \pi$.

4.4. Find the exact solution to question 1.4. given the initial condition $y(2) = 3$.

Q5

5.1 Find the general solution to $\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{5}{x}y = 3x + 2$.

5.2 Find the general solution to $\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{-4y}{4x+1} = 5x^4$.

5.3 Find the exact solution to $\frac{\mathrm{d}y}{\mathrm{d}x} - \cos(x) = -4y\cot(x)$ when $y(\frac{\pi}{2}) = 1$.

5.4 Find the exact solution to $e^{4x}\frac{\mathrm{d}y}{\mathrm{d}x} + 4e^{4x}y = 4x\sin(x)$ when $y(0) = 4$.

After attempting the questions above, please click this link to find the answers.

Version history and licensing

v1.0: initial version created 04/06 by mak.